

Derivation: Exact Edge-Compatibility Constraints on W

Grossman & Boudaoud, PRR 2026, arXiv:2309.07844 — self-contained working notes

1. Setup and Notation

1.1 The mesh

Consider a 2D elastic spring network on a triangulated domain with:

- n vertices in 2D $\rightarrow 2n$ positional degrees of freedom
- N triangles, E_{int} interior edges (shared by two triangles), E_{bd} boundary edges

By Euler's formula for a simply-connected triangulated domain ($n - E + N = 1$), combined with $3N = 2E_{\text{int}} + E_{\text{bd}}$:

$$N = 2n - n_{\text{bd}} - 2 \approx 2n, \quad E_{\text{int}} = 3n - 2n_{\text{bd}} - 3 \approx 3n, \quad E_{\text{total}} = E_{\text{int}} + E_{\text{bd}} \approx 3n$$

for large meshes where $n_{\text{bd}} \sim \sqrt{n} \ll n$.

1.2 Metrics and the non-affine response tensor W

Each triangle s carries a symmetric 2×2 metric tensor $g(s)_{\mu\nu}$ (3 independent components). When a macroscopic metric perturbation $\Delta g_{\alpha\beta}$ is applied, the local response at triangle s is:

$$g(s)_{\mu\nu} = \bar{g}_{\mu\nu} + \Delta g_{\mu\nu} + \delta g(s)_{\mu\nu}$$

where $\bar{g}$ is the reference (rest) metric, Δg is the uniform affine part, and $\delta g(s)$ is the **non-affine fluctuation** specific to triangle s .

Because the system is linear in Δg , we write:

$$\delta g(s)_{\mu\nu} = W(s)^{\alpha\beta}_{\mu\nu} \Delta g_{\alpha\beta}$$

The tensor $W(s)^{\alpha\beta}_{\mu\nu}$ is the **non-affine response tensor**: its upper index $\alpha\beta$ labels the loading direction (which component of Δg is being applied), and its lower index $\mu\nu$ labels the resulting local metric deviation. It has 9 independent components per triangle (3 lower $\times$ 3 upper, using the symmetry $\alpha\beta = \beta\alpha$, $\mu\nu = \nu\mu$).

Goal: find $W(s)^{\{\alpha\beta\}}_{\{\mu\nu\}}$ for all triangles ss and all indices.

2. The Energy

Each triangle ss has a per-triangle elastic stiffness tensor $A(s)^{\{\mu\nu\alpha\beta\}}$, with the major symmetry $A^{\{\mu\nu\alpha\beta\}} = A^{\{\alpha\beta\mu\nu\}}$ (from its definition as a second derivative of energy, and from the symmetry of W — both established in the paper).

The elastic energy stored in triangle ss is the squared departure of its local metric from its rest metric $\bar{g}$, weighted by $A(s)$. The total metric departure is $\Delta g + \delta g(s)$, so:

$$E = \frac{1}{2} \sum_s A(s)^{\{\mu\nu\alpha\beta\}} \bigl(\Delta g^{\{\mu\nu\}} + \delta g(s)^{\{\mu\nu\}} \bigr) \bigl(\Delta g^{\{\alpha\beta\}} + \delta g(s)^{\{\alpha\beta\}} \bigr)$$

We treat Δg as fixed (the applied loading) and minimize over $\delta g(s)$ subject to constraints.

Define $\delta A(s) := A(s) - \langle A \rangle$, the deviation of triangle ss 's stiffness from the spatial mean $\langle A \rangle = \frac{1}{N} \sum_{s'} A(s')$. For a uniform network $\delta A(s) = 0$; heterogeneity enters through δA .

3. Constraints

3.1 Global normalization constraint (χ)

The non-affine fluctuations $\delta g(s)$ must average to zero — this is the **definition** of the split between affine (Δg , the macroscopic part) and non-affine ($\delta g(s)$, the fluctuation):

$$\sum_s \delta g(s)^{\{\mu\nu\}} = 0 \quad \text{for each } \{\mu\nu\}$$

This is **3 scalar equations** (one per independent symmetric-tensor component $\{\mu\nu\} \in \{00, 01, 11\}$). It contains the physical area-conservation condition $\bar{g}^{\{\mu\nu\}} \sum_s \delta g^{\{\mu\nu\}}(s) = 0$ as a special case (the trace), plus two further normalization conditions fixing the shear components of the mean.

Enforced by a **Lagrange multiplier** $\chi^{\{\mu\nu\}}$ (a symmetric 2×2 tensor, 3 independent components). The Lagrangian term is:

$$\chi^{\{\mu\nu\}} \sum_s \delta g(s)^{\{\mu\nu\}}$$

Note: $\chi^{\{\mu\nu\}}$ is "really" a single scalar χ times $\bar{g}^{\{\mu\nu\}}$ only for the area-conservation part. The full 3-component $\chi^{\{\mu\nu\}}$ is

required to normalize all components of $\langle \delta g \rangle$, not just the trace. Taking $\partial \delta g(s)_{\mu\nu} / \partial \chi^{\mu\nu}$ contributes $+\chi^{\mu\nu}$ to every triangle.

3.2 Edge compatibility constraint (λ)

Two adjacent triangles s_1 and s_2 sharing a reference edge Δx^μ must agree on that edge's length:

$$g(s_1)_{\mu\nu} \Delta x^\mu \Delta x^\nu = g(s_2)_{\mu\nu} \Delta x^\mu \Delta x^\nu$$

Since $\bar{g}$ is shared, this reduces to:

$$\bigl[\delta g(s_1)_{\mu\nu} - \delta g(s_2)_{\mu\nu} \bigr] \Delta x^\mu \Delta x^\nu = 0$$

This is **one scalar equation per interior edge** $e = (s_1, s_2, \Delta x)$.

Enforced by a **Lagrange multiplier** λ_e (one scalar per interior edge). The Lagrangian term is:

$$\lambda_e \bigl[\delta g(s_1)_{\mu\nu} - \delta g(s_2)_{\mu\nu} \bigr] \Delta x^\mu \Delta x^\nu$$

Taking $\partial \delta g(s)_{\mu\nu} / \partial \chi^{\mu\nu}$, the contribution to triangle s is:

$$+\lambda_e \Delta x^\mu \Delta x^\nu \quad (s = s_1), \quad -\lambda_e \Delta x^\mu \Delta x^\nu \quad (s = s_2)$$

This has the structure: **one scalar** λ_e times the **fixed rank-1 reference tensor** $\Delta x^\mu \Delta x^\nu$. This is exactly analogous to the χ term ($\chi^{\mu\nu} = \text{scalar} \times \text{reference tensor } \bar{g}^{\mu\nu}$), with the difference that $\Delta x \otimes \Delta x$ is rank-1 (not invertible), so λ_e cannot be promoted to a full symmetric tensor — it stays a scalar per edge.

4. The Full Lagrangian and Stationarity

$$\mathcal{L} = E + \chi^{\mu\nu} \sum_s \delta g(s)_{\mu\nu} + \sum_{e=(s_1, s_2, \Delta x)} \lambda_e \bigl[\delta g(s_1)_{\mu\nu} - \delta g(s_2)_{\mu\nu} \bigr] \Delta x^\mu \Delta x^\nu$$

Taking $\partial \mathcal{L} / \partial \delta g(s)_{\mu\nu} = 0$, using the major symmetry of $A(s)$ so that both cross terms combine, gives the **per-triangle stationarity**:

$$\sum_s A(s) (\Delta g + \delta g(s)) = \langle A \rangle_N (\Delta g + \langle \delta g \rangle) + \chi^{\mu\nu} + \sum_{\langle e \rangle} \mathrm{sign}(s,e) \lambda_e \Delta x_e^\mu \Delta x_e^\nu = 0 \quad (*)$$

where $\mathrm{sign}(s,e) = +1$ if $s = s_1$, -1 if $s = s_2$, and the sum runs over all interior edges incident to triangle s .

5. Eliminating χ (Sum Over All Triangles)

Sum $(*)$ over all $s = 1, \dots, N$:

- **A-term:** $\sum_s A(s) (\Delta g + \delta g(s)) = \langle A \rangle_N (\Delta g + \langle \delta g \rangle)$
- **χ -term:** $N \chi^{\mu\nu}$ (same at every triangle)
- **λ -terms:** each interior edge $e = (s_1, s_2)$ contributes $+\lambda_e$ at s_1 and $-\lambda_e$ at s_2 — they **telescope to zero**

Result:

$$\langle A \rangle^{\mu\nu} (\Delta g + \langle \delta g \rangle) + \chi^{\mu\nu} = 0$$

Applying the normalization constraint $\langle \delta g \rangle = 0$:

$$\chi^{\mu\nu} = -\langle A \rangle^{\mu\nu} \Delta g$$

Key observation: χ is fully determined by Δg and the mean stiffness $\langle A \rangle$. The λ terms leave no trace in this global equation — they cancel pairwise across each shared edge. The global constraint $\langle \delta g \rangle = 0$ is what closes the equation for χ ; without it, χ would be underdetermined.

6. Substituting χ Back: The Compact Equation

Substitute $\chi = -\langle A \rangle \Delta g$ into $(*)$:

$$A(s) (\Delta g + \delta g(s)) - \langle A \rangle (\Delta g + \sum_{\langle e \rangle} \mathrm{sign}(s,e) \lambda_e \Delta x_e^\mu \Delta x_e^\nu) = 0$$

where $q_e^{\mu\nu} = \Delta x_e^\mu \Delta x_e^\nu$ (the rank-1 carrier tensor for edge e).

$$A(s) \delta g(s) + \delta A(s) \Delta g - \langle A \rangle \delta g + \langle A \rangle \langle \delta g \rangle + \sum_{\langle e \rangle} \mathrm{sign}(s,e) \lambda_e q_e^{\mu\nu} = 0$$

Using $\langle \delta g \rangle = 0$ to simplify $\langle A, \delta g \rangle$:

$$\langle A, \delta g \rangle = \langle (\delta A + \langle A \rangle) \delta g \rangle = \langle \delta A, \delta g \rangle + \langle A \rangle \underbrace{\langle \delta g \rangle}_{=0} = \langle \delta A, \delta g \rangle$$

Define the operator B acting on a field $x(s)$ as:

$$(Bx)(s)^{\mu\nu} := \frac{1}{N} \sum_{s'} \delta A(s')^{\mu\nu\alpha\beta} x(s')_{\alpha\beta}$$

(same $\delta A(s')$ for all s , determined by the **column** triangle s' — this is the structure in the existing code). Then the per-triangle equation becomes:

$$[A(s) - B] \delta g(s) + \sum_{e \ni s} \text{sign}(s, e) \lambda_e q_e = -\delta A(s), \Delta g \tag{*}$$

In compact matrix-vector notation over all triangles:

$$\boxed{(A - B) \delta g + J^{\text{top}} \lambda = -\delta A, \Delta g}$$

where: $(A - B)$ is the $3N \times 3N$ mean-field operator (block-diagonal A , rank-3N correction B coupling all triangles via column- δA) - J is the $E_{\text{int}} \times 3N$ edge-constraint matrix: row e has $+q_e^{\text{top}}$ at the 3 columns of s_1 and $-q_e^{\text{top}}$ at the 3 columns of s_2 - $\lambda \in \mathbb{R}^{E_{\text{int}}}$ is one scalar multiplier per interior edge - RHS: $-\delta A(s), \Delta g$ is the 3-component driving vector at triangle s

Setting $J=0$ (no edge constraints): $(A - B) \delta g = -\delta A, \Delta g$, which is exactly the paper's Eq. 19 before stripping Δg .

7. Stripping Δg : The Equation for W and Λ

The system $(*)$ is linear in Δg (the RHS is $\propto \Delta g$, the LHS is $\propto \delta g$ and $\propto \lambda$, both of which are also linear in Δg by linearity of the constrained solution map). Therefore:

$$\delta g(s)^{\mu\nu} = W(s)^{\alpha\beta\mu\nu} \Delta g_{\alpha\beta}, \quad \lambda = \Lambda^{\alpha\beta} \Delta g_{\alpha\beta}$$

where W and Λ are Δg -independent coefficient tensors. Substituting into $(*)$ and stripping $\Delta g_{\alpha\beta}$ (valid since it must hold for all independent components of Δg):

$$\boxed{(A - B)W + J^{\text{top}} \Lambda = -\delta A}$$

together with the stripped constraint $JW = 0$.

This is the **final equation for W and Λ** — loading-independent. The full $9N \times 9N$ system (9 components of W per triangle, Λ of size $3 \times E_{\text{int}}$):

$$\begin{pmatrix} A-B & J^{\text{top}} \\ J & 0 \end{pmatrix}$$

$$\begin{pmatrix} W \\ \Lambda \end{pmatrix}$$

$$-\Delta A \end{pmatrix}$$

Here: - W is $9N$ unknowns (the full non-affine response tensor, all loading and metric components) - Λ is $3 \times E_{\text{int}}$ unknowns (one loading-indexed scalar per interior edge; $\Lambda^{\alpha\beta}$ is the coefficient of $\Delta g^{\alpha\beta}$ in Λ) - J is now $3E_{\text{int}} \times 9N$: block-diagonal in loading, $J = I \otimes J_0$ with $J_0 \in \mathbb{R}^{E_{\text{int}} \times 3N}$ (same edge geometry for each loading)

When $J=0$ this reduces to $(A-B)W = -\Delta A$, i.e., the paper's Eq. 19-20. ✓

Note on symmetry of $A-B$. The code writes $B_{IJ} = \frac{1}{N} \Delta A(s')$ (column triangle s'), which is not manifestly symmetric. However, $(A-B)$ is the Hessian of the quadratic form $\mathcal{L}[W, \chi]$ in W after integrating out χ (a Schur complement of a symmetric matrix), so it must be symmetric. The apparent asymmetry is an artifact of the substitution step; the underlying variational structure guarantees $A-B = (A-B)^{\text{top}}$. This is stated explicitly rather than used silently.

8. Closed-Form Solution via Woodbury

Let $P := A - B$ and $W_0 := -P^{-1} \Delta A$ (the **existing unconstrained solution**, computed by the current Woodbury code, paper Eq. 20). The KKT system is solved as:

1. **Unconstrained solve** (existing code): $W_0 = -P^{-1} \Delta A$
2. **Constraint violation**: $r = J, W_0 \in \mathbb{R}^{3E_{\text{int}}}$ — how much W_0 violates edge compatibility
3. **Gram matrix** $G = J, P^{-1} J^{\text{top}} \in \mathbb{R}^{3E_{\text{int}} \times 3E_{\text{int}}}$: each row of J is a sparse $9N$ -vector; applying P^{-1} to it reuses the same Woodbury factors (A^{-1}, S, z) already computed in step 1.
4. **Multipliers**: $G \Lambda = r$ (dense solve, size $3E_{\text{int}} \approx 4.5N$)
5. **Correction**: $W = W_0 - P^{-1} J^{\text{top}} \Lambda$

Geometrically: W is the **projection of W_0 onto $\ker J$** in the P -metric — the closest compatible field to the unconstrained mean-field solution.

9. Degrees of Freedom

Mesh sizes (per loading, i.e., for one $\alpha\beta$ slice of W):

quantity	formula	$\approx$ large mesh
triangle metric components (unknowns)	$3N$	$6n$
interior edge constraints	E_{int}	$3n$
global normalization $\langle \Delta g \rangle = 0$	3	3
residual DOF	$3N - E_{\text{int}} - 3$	$3n - 3$
physical vertex DOF	$2n$	$2n$

The residual $3n > 2n$: edge-length constraints are **weaker** than full metric compatibility. Each edge constraint enforces agreement on the rank-1 projection $q_{\mu\nu} = \Delta x^\mu \Delta x^\nu$, not on all 3 components of the metric. Full St. Venant compatibility (flatness) would additionally require the Gaussian curvature deficit at each interior vertex to vanish — roughly $n_{\text{int}} \approx n$ extra conditions — which would reduce the count to $\sim 2n$. The spring network imposes only edge-length compatibility, which is the physically correct constraint for a system whose springs constrain edge lengths, not the full strain tensor.

For the full W (all loadings):

quantity	count
unknowns W	$9N \approx 18n$
multipliers Λ	$3E_{\text{int}} \approx 9n$
stationarity equations	$9N \approx 18n$
constraint equations	$3E_{\text{int}} \approx 9n$

Augmented system: exactly square ($9N + 3E_{\text{int}}$) in both rows and columns. ✓

J has full row rank $3E_{\text{int}}$. $J = I_3 \otimes J_0$; for $J_0 \in \mathbb{R}^{E_{\text{int}} \times 3N}$, the null space is all compatible metric fields (parametrized by one length per edge): $\dim \ker J_0 = E_{\text{total}}$, so $\text{rank}(J_0) = 3N - E_{\text{total}} = E_{\text{int}}$, hence $\text{rank}(J) = 3E_{\text{int}}$. No over-determination. ✓

Multipliers add no net DOF: each Λ entry is paired with one constraint equation, keeping the augmented system square. Physical DOF drop from $9N$ to $3E_{\text{total}}$ (the compatible subspace of the metric field).

10. Files to Modify (Implementation — not yet started)

File	Change
Phase 2/ forward_solver_torch.py	(1) <code>build_triangle_adjacency()</code> returning interior edge pairs and Δx vectors; (2) augment <code>_woodbury_solve()</code> to accept adjacency and execute steps 2-5 of §8; (3) cache adjacency in <code>ElasticSolver.__init__</code>
Phase 4/gnn/train.py	Pass adjacency to solver at construction — no new hyperparameter

11. Implementation Steps

- Backup:** copy `Phase 2/forward_solver_torch.py` → `Phase 2/forward_solver_torch.py.bckp` and `Disc_2_Cont_optimized.py` → `Disc_2_Cont_optimized.py.bckp`.
- Add `build_triangle_adjacency()`** to `ElasticSolver`: iterate over triangle pairs, find shared edges, record $(s_1, s_2, \Delta x)$ for each interior edge. Cache as buffers in `__init__`. Build $J_0 \in \mathbb{R}^{E_{\text{int}} \times 3N}$ with rows $q_e = (dx^2, 2dx dy, dy^2)$ at columns of s_1 and $-q_e$ at columns of s_2 . Full $J = I_3 \otimes J_0$ as a sparse $3E_{\text{int}} \times 9N$ matrix.
- Augment `_woodbury_solve()`**: after computing W_0 (existing 5 lines), add the 5-step KKT correction (§8): compute $r = J W_0$, build $G = J P^{-1} J^{\text{top}}$ by applying the Woodbury solve to each row of J (reuse A_{inv} , SS , z), solve $G \Lambda = r$, correct $W = W_0 - P^{-1} J^{\text{top}} \Lambda$.
- Pass adjacency** from `ElasticSolver.__init__` to `_woodbury_solve` — no new public API, no hyperparameters; constraints are hard (always on).

12. Verification Plan

Step 0 — Backup

```
cp "Phase 2/forward_solver_torch.py" "Phase 2/forward_solver_torch.py.bckp"
cp "Disc_2_Cont_optimized.py" "Disc_2_Cont_optimized.py.bckp"
```

Step 1 — Uniform mesh (sanity)

All springs identical $\rightarrow \Delta A(s)=0$ for all $s \rightarrow W_0=0 \rightarrow r=J W_0=0 \rightarrow \Lambda=0 \rightarrow W=0$. Verify numerically that the solver returns $W=0$ and $\nu = \nu_{\text{uniform}}$ matches the analytic value for the regular triangular lattice ($\nu = 1/3$).

Step 2 — Crystalline and hexagonal tiling (regression)

Run the new solver on the existing crystalline (foam_eta00) and regular hexagonal lattice configs used in past sessions. Compare ν , E , full elastic tensor C against the values recorded in Phase 3/results_summary.txt . Expect <1% change for low-contrast cases (mean-field was already accurate there).

Step 3 — Auxetic response vs η (key test)

For the perturbed hexagonal lattice with perturbation parameter $\eta \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$: - Run **old** solver (from .bckp) and record $\nu(\eta)$. - Run **new** solver and record $\nu(\eta)$. - Plot both curves. Expect the new solver to show stronger auxetic (more negative) ν at high η , correcting the mean-field over-estimate. Compare against any available simulation data in Phase 3/ or ANALYTICAL_MODEL_STATUS.md .

Step 4 — Binary-contrast validation

For the VD rigidity pattern (the case with ~70% error documented in ANALYTICAL_MODEL_STATUS.md): compare old vs new ν against the KUBC simulation value. Expect significant improvement toward the simulation value.

Step 5 — Gradient check

torch.autograd.gradcheck on the full augmented solve (small mesh, double precision) to confirm end-to-end differentiability w.r.t. rigidities and rest lengths — required for Phase 3/4 inverse optimization to remain valid.